MIL-B-20278B (NAVY)
22 April 1953

SUPERSEDING
MIL-B-20278A(BuOrd)
21 October 1952
MIL-B-20278
9 November 1951

MILITARY SPECIFICATION

BLADDERS, FLOTATION

(All interested Bureaus of the Navy Department have concurred in the use of this specification for purchase.)

1. SCOPE

1.1 Scope. - This specification covers flotation bladders to provide bouancy for personnel carrying equipment (Type 1) and for various demolition outfits (Type 2).

1.2 Classification. - Flotation bladders shall be of the following types, as specified (see 6.1):

Type 1 - With straps
Type 2 - Without straps

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

V-T-276 - Thread, Cotton.
CCC-T-191 - Textile Test Methods.
UU-P-268 - Paper, Kraft, Wrapping.
UU-P-271 - Paper, Kraft, Wrapping, Waterproof.
CCC-T-191 - Textile Test Methods.
LLL-B-631 - Boxes, Fiber, Corrugated (For Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid (For Domestic Shipment).

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MILITARY

JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN-W-530 - Webbing, Cotton, Natural or in Colors.
MIL-T-3530 - Treatment, Mildew-Resistant, for Thread and Twine.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

ARMY

6-187 - Cloth, Cotton, Olive-Drab, for Bandoliers.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking of Shipments.

DRAWINGS

BUREAU OF ORDNANCE

Type 1 (with straps) - Drawing No. 982514 (Figure 1)
Type 2 (without straps) - Drawing No. 982515 (Figure 2)

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 - General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance equipment.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS.

3.1 Materials. - All materials used in the manufacture of the flotation bladders shall conform strictly to the drawings and specifications unless approval for a departure therefrom has been obtained from the procuring agency. When alternate materials or methods of manufacture are permitted, the bidder's selection shall be clearly stated in the proposal. The use of materials or methods of manufacture differing from those indicated on the drawings will be considered when such changes are advantageous to the Government.

3.1.1 Mildewproofing. - The thread shall be mildew-proofed according to Specification MIL-T-3530, Class 2. The cloth and the cotton webbing shall be mildew-proofed by application of a solvent solution of 1.00 to 2.00 percent dihydroxy dichloro diphenyl methane.

3.1.2 Cloth. - The cloth shall be bandolier sheeting, vat dyed Olive Drab No. 7. The finished cloth shall conform to the requirements of Army Specification 6-187, including the following:

Weave	Weight per Square Yard, Ounces, Min.	Thread Count, Per Inch, Minimum		Breaking Strength (Grab Method), Pounds, Minimum	
		Warp	Filling	Warp	Filling
Plain	4.6	68	68	70	70

3.1.3 Webbing. - The webbing, for body straps on Type 1 bladder only, shall be 3/4 inch wide, vat dyed Olive Drab No. 7, and shall conform to the requirements of Specification JAN-W-530, Type I. The Type 1 bladder shall be provided with one neck strap and one body strap. Each strap shall be cut 42 inches long with one end looped through the grommet and securely stitched with double box-stitching. The second strap shall be placed diagonally opposite the eyelet to which the first strap is fastened. Free ends of straps shall be dipped in paraffin to prevent raveling. Each strap shall measure, after overlapping and stitching, not less than 36 inches long.

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3.1.4 Thread. - The thread shall be No. 20, 4-cord, vat dyed Olive Drab No. 7 cotton thread, silk finish, left twist, conforming to Type IB3 of Specification V-T-276.

3.1.5 Coating compound. - The coating compound shall be a suitably polymerized or co-polymerized vinyl resin, synthetic rubber, or new natural rubber, suitable pigmented and plasticized. The compound used shall be non-toxic and non-irritating and shall contain no water soluble material. Thermoplastic polyvinyl butyral is not permitted. If thermosetting polyvinyl butyral is used, it shall be so compounded and modified as to permit the material to undergo thermosetting or heat conversion; and after conversion, the finished coating shall not have thermoplastic properties and shall not be sensitive to ethanol. Bladders made of natural rubber, thermosetting polyvinyl butyral or other synthetic rubber shall be given a heat or curing treatment after assembly. The coating compound shall be applied to one side of the cloth only. The compound shall have good adhesion to the cloth, shall be flexible, free of tackiness, resistant to abrasion and scratching and shall not tender the fabric. The coated fabric shall be free of pinholes, relatively free of windows, etc., shall conform to the requirements of Table I and those in Section 4 of this specification, and shall have a weight per square yard between 5 ounces minimum and 8 ounces maximum.

3.1.6 Cement. - The cement shall have good adhesion, flexibility, and all other characteristics equal to the requirements for the coating of the fabric. Natural or reclaimed rubber shall not be used in the cement or sealing compound. Joinings, made by heat sealing or by solvent action alone shall not be permitted. The cement shall be so compounded as to be heat-curing or thermosetting.

3.1.7 Valve tube assembly. - Shall be made of high grade synthetic rubber, 1/2 ± 1/16 inches outside diameter, with a wall thickness of 3/32 ± 1/64 inches. The valve tube shall be moulded to an oval-shaped flange. The flange shall measure approximately 1-3/4 inches wide by 2-1/4 inches long. The assembly shall be inserted through the coated fabric with the flange well cemented to the inside of the bladder as shown in the drawing.

3.1.8 Keeper. - A keeper for the valve tube, made of a strip of the coated fabric, double textured, shall be double box-stitched to the upper seam of the bladder. The keeper shall be cut 4-1/8 inches long and shall finish 1/2 inch wide with a loop not less than 1-1/2 inches.

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3.1.9 Grommets. - The grommets, for Type 1 Bladders only, shall be of non-corrosive metal with a dull finish and shall be 5/16 inch inside diameter and 1/2 inch outside diameter.

3.2 Design and Construction. - The design and construction shall be as shown in Bureau of Ordnance Drawing No. 982514 for Type 1, and Drawing No. 982515 for Type 2 (Figures 1 and 2, respectively). All parts requiring cementing shall be laid in cement and rolled so as to secure good adhesion and air-tight seams and joinings. The seams shall be not less than 3/4 inch wide. A reinforcement stay cut 1 inch wide made of the same material as the bladder, shall be cemented at the joining of the seams as shown in the drawings.

3.3 Before production is begun, unless otherwise specified, two samples of the finished bladder shall be submitted to the procuring agency.

3.4 Completeness. - Each flotation bladder shall be delivered completely assembled, marked, tested, passed by the Government Inspector, packaged and packed in accordance with the requirements of this specification.

3.5 Manufacturing accessories. - The manufacturer shall provide all tools to be used in the manufacture of the flotation bladders.

3.6 Weight. - The weight of the finished bladder shall not exceed 7 ounces.

3.7 Tolerances. - Tolerances in size will be as shown on the drawings and within the limits of good manufacturing practice.

3.8 Inflation requirements. - The finished inflated bladders shall be capable of passing the tests specified in 4.6.13 and 4.6.14.

3.9 Water contamination. - The bladder shall be suitable for use as a drinking-water container. Water contained for 24 hours shall not be contaminated or adversely affected in taste or odor.

3.10 Marking. - Each bladder shall be marked, clearly and legibly with black water-proof ink, as shown on the drawing.

3.11 Workmanship. - The bladders shall be clean, well finished and free from perforations or other imperfections that might affect their appearance or serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

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4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specification for Inspection of Material and Bureau of Ordnance Pamphlet OP-400.

4.2 Sampling. - In determining acceptability of the flotation bladders the Government Inspector will use sampling plans and procedures, when applicable, conforming to the provisions of MIL-STD-105.

4.3 Inspection lot. - An "inspection lot" shall mean a collection of flotation bladders submitted for Government inspection. Unless otherwise specified the number of flotation bladders in an inspection lot shall be determined by the Government Inspector and may differ from the quantity designated in the contract or order as a lot for production or other purposes.

4.4 Contractor's inspection. - The contractor shall inspect all finished bladders for compliance with this specification prior to submission to the Government Inspector for final inspection.

4.5 Government inspection. - During the process of manufacture and assembly, the Government Inspector shall perform, witness or participate in such inspections and tests as are required by Acceptance Inspection Instructions furnished by the procuring agency, and as otherwise considered necessary to assure conformance of component parts, sub-assemblies and assemblies with applicable contract, order, requisition, drawing or specification requirements. The manufacturer shall provide the Government Inspector with all facilities necessary for the performance of his work.

4.6 Tests. - The following tests of samples of the cured coated fabric shall be conducted according to the requirements of Specification CCC-T-191, except as stated herein. In addition, tests of samples of the cured coated fabric having finished seams shall be conducted according to the requirements of 4.6.5, 4.6.6, 4.6.8 and 4.6.9. Specific requirements shall be as given in Table I.

4.6.1 Tearing strength test. - The tearing strength test shall be made on an Elmendorf Testing Machine with augmenting attachment, according to Method 5132, Specification CCC-T-191. Report scale reading of tear resistance across warp and filling threads.

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4.6.2 Hydrostatic test. - Hydrostatic pressure shall be applied to the coated side of the coated fabric by insertion in a Mullen or similar testing machine. To adapt the Mullen tester, the diaphragm is removed and water used as the liquid. The water level should be flush with the surface of the tester so that no air pocket is formed. Apply the pressure slowly and steadily at a rate of speed not exceeding 1/2 revolution per second. The pressure reading is taken at the first appearance of water through the fabric being tested. See Method 5512, Specification CCC-T-191, for details of test.

4.6.3 Blocking test. - This test shall be conducted according to Method 5872, Specification CCC-T-191.

4.6.4 Flexibility. - Four 1 inch x 10 inch strips of the coated fabric shall be taken for test. Two strips shall be cut parallel to the warp threads of the material and two strips parallel to the filling threads of the material. The test strips shall be conditioned for 3 hours in a room maintained at a temperature of 70° ± 4°F., and 65 percent relative humidity. The test shall be conducted according to Specification CCC-T-191, Method 5200, using the apparatus shown therein. The flexibility reading shall be taken one minute after the sample is placed in position. The average of the vertical distance from the top of the bar to the bottom of the loop shall constitute the flexibility.

4.6.5 Flexibility after heat treatment. - This test shall be conducted according to Specification CCC-T-191, Method 5870. The samples are to be tested as described in 4.6.4 before performing this test.

4.6.6 Flexibility after water leaching. - This test shall be conducted according to Specification CCC-T-191, Method 5830. Four samples are to be prepared as described in 4.6.4, and the original flexibility determined, before proceeding with this test.

4.6.7 Natural weathering test. - The coated fabric, after unprotected exposure to the elements for 30 days, shall show no appreciable change in color, pliability or strength, and shall not show signs of flaking, peeling, checking, cracking or loss of coating.

4.6.8 Weatherometer test. - The coated side of the fabric shall be exposed and subjected to 100 hours in the Twin-Arc-Weatherometer. The temperature of the atmosphere inside the

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Weatherometer, around the exposed samples shall be 125° ± 5°F. After this exposure the sample shall have lost not more than 25 percent of its original hydrostatic resistance determined in accordance with 4.6.2, nor shall there be any appreciable stiffening of the sample.

4.6.9 Aging test. - This test shall be performed in accordance with Specification CCC-T-191, Method 5852.

4.6.10 Adhesion test. - This test is to determine the resistance to separation of the coating from the cloth. The test shall be conducted according to Method 5970, Specification CCC-T-191.

4.6.10.1 Dry test on seams. - The adhesion of the seams of the bladder (including that of the reinforcement stay) shall be tested by separating for a distance of three inches and inserting in the jaws of the testing machine. The test shall be conducted as stated in 4.6.10.

4.6.10.2 Water test on seams. - The seams, after two hours immersion in water at 75° ± 2°F., and tested while wet, shall meet the requirements of 4.6.10.1.

4.6.11 Strength of coating. - A test piece, 6 X 6 inches, shall be cut from the coated fabric in a direction 45 degrees to the warp or filling threads and clamped in jaws two inches wide and three inches apart on a standard testing machine (Scott or equal). The sample shall be pulled apart at a constant rate of two inches per minute until a total load of 30 pounds has been applied. The load shall be applied for 30 seconds. The test shall be repeated with duplicate pull at a 90 degree angle from the first.

4.6.12 Folding tube at low temperature. - The tube shall be subjected to a temperature of minus 20°F. for 30 minutes, then folded upon itself and pressed lightly between the fingers at a point 1-1/4 inches from the fold, so as to effect contact at the point of pressing. The temperature of the tube shall be maintained at minus 20°F. throughout the test.

4.6.13 Inflation test. - Bladders shall be inflated to 1-1/2 lbs. per square inch air pressure. The tube shall be properly folded, inserted through the keeper and well stoppered with a suitable plug. Bladders shall remain inflated without loss of air pressure for 12 hours. Bladders showing a loss of air pressure shall be subjected to the water immersion test specified in 4.6.14. There shall be no tearing or separation of the seams.

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4.6.14 Water immersion test. - Bladders shall be inflated to 1 lb. per square inch air pressure. The tube shall be properly folded, inserted through the keeper and well stoppered with a suitable plug. Upon immersion of the bladders in water there shall be no visible leakage of air. (Bubbles or entrapped air adhering to the outside of the bladder should be ignored.) There shall be no tearing or separation of the seams.

4.6.15 Breaking strength of coated fabric. - The coated fabric shall be tested for breaking strength in accordance with the requirements of Specification CCC-T-191, Method 5100.

5. PREPARATION FOR DELIVERY.

5.1 Packaging. - Each bladder shall be folded in 3 equal sections of its length, and the inflator tube of one bladder shall be threaded through the keeper strap of another to form a unit of 2 bladders.

5.2 Packing. - The bladders shall be packed in overseas or domestic shipping containers, as specified in the invitation for bids. (See 6.1).

5.2.1 For domestic shipment.- Unless otherwise specified, fifty units (100 bladders) shall be packed in corrugated fiber or solid fiber boxes conforming to specifications LLL-B-631 or LLL-B-636, respectively. Containers shall be lined with No. 2 Kraft wrapping paper as specified in 5.2.2.

5.2.2 For overseas shipment. - Fifty units (100 bladders) shall be packed in a V3C or V3S shipping container conforming to the requirements of specification JAN-P-108. Containers shall be FOL style (outside flaps full overlap, inside flaps same length as outside). Containers shall be lined with Kraft paper conforming to Specification UU-P-271 Type C, except that the strength of the paper shall be determined by a Mullen test and shall have a value of not less than 48 points, and the temperature for the bleeding test shall be 170°F. Each container shall be interlined with No. 2 Kraft wrapping paper conforming to Specification UU-P-268 (60 lbs. basis, 24 X 36 inches, 500 sheets). Paper used in lining boxes shall be of sufficient size to overlap at all edges not less than 3 inches, including fold down.

5.3 Marking. - In addition to special marking required by the contractor order interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

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5.3.1 The nomenclature of these flotation bladders is:

Flotation Bladder Mk 1 Mod 0 or Flotation Bladder Mk 1 Mod 1 as specified in the contract or order.

5.3.2 The stock (or code) numbers of the flotation bladders are:

Flotation Bladder Mk 1 Mod 0, Type 1:

Navy - 072-1520-E
Marine Corps - 74-B-295

Flotation Bladder Mk 1 Mod 1, Type 2:

No number assigned.

6. NOTES

6.1 Ordering Data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (See 1.1).
(c) Whether domestic or overseas packing is required (See Section 5).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Navy-Bureau of Ordnance

Other interest:
Navy-MCShS

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TABLE I

TEST REQUIREMENTS

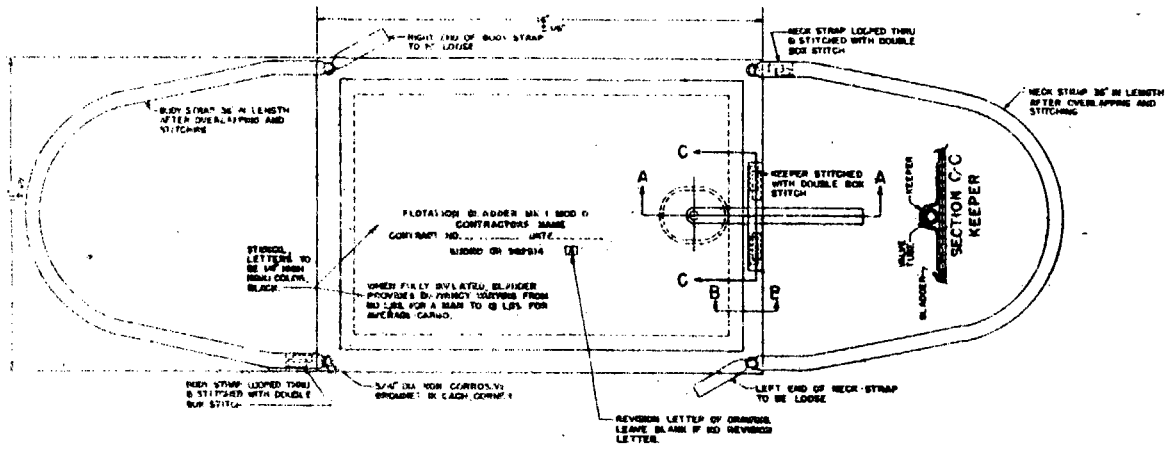
Paragraph	Kind of Test	Requirements
4.6.1	Tearing Strength Test:	Warp 25 Filling 20 (Scale Reading)
4.6.2	Hydrostatic Test:	40 Pounds (Minimum)
4.6.3	Blocking Test:	No adhesion or damage of surface
4.6.4	Flexibility Tests:	Determine initial flexibility
4.6.5	After Heat Treatment	Maximum Loss in Flexibility: 7-1/2% from initial reading
4.6.6	After Water Leaching	Maximum Loss in Flexibility: 7-1/2% from initial reading
	Weathering Tests:	
4.6.7	Natural Weathering	No appreciable change in color, pliability or strength and no sign of flaking, peeling, checking, cracking or loss of coating.
4.6.8	Weatherometer	Maximum loss of hydrostatic resistance: 25% No appreciable stiffening of sample.
4.6.9	Aging Test	Maximum loss in tensile strength: 25% Show no sign of becoming stiff and brittle or soft and tacky.

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TABLE I (Contd)

Paragraph	Kind of Test	Requirements
	Adhesion Tests:	
4.6.10	Coating	Minimum Pull: 7 lbs.
4.6.10.1	Seams (Dry)	Minimum Pull: 1-1/2 lbs.
4.6.10.2	Seams (After Water Test)	Minimum Pull: 1-1/2 lbs.
4.6.11	Strength of Coating:	Maximum loss in hydrostatic resistance (4.6.2):50% Coating shall not break or crack.
4.6.12	Folding Tube at low temperature after folding.	shall not crack at minus 20°F. nor leak
4.6.13	Inflation Test	Remain inflated for 12 hrs. without loss in pressure; no tearing or separation of seams.
4.6.14	Water Immersion Test	No visible leakage during immersion; no tearing or separation of seams.
4.6.15	Breaking Strength of Coated Fabric (1" X 1" X3" Grab Method)	Warp, 55; Filling, 55 (Pounds minimum)

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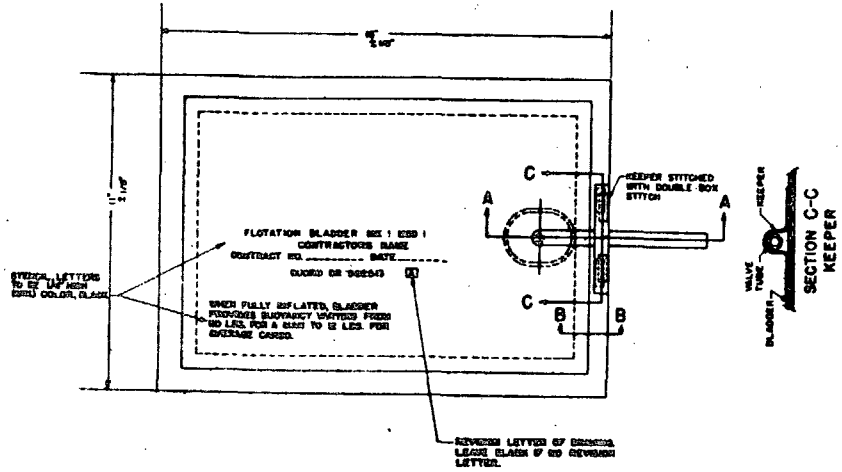
REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	CANCELLED NO NOTE	3/16/70	

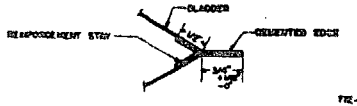
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		FLOTATION BLADDER MK I MOD O	DEPARTMENT OF THE NAVY BUREAU OF SHIPS
			USCGD 982514

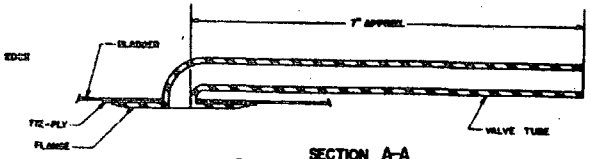
FIGURE 1

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SECTION B-B
DETAIL OF
REINFORCEMENT STAY



SECTION A-A
(A) VALVE TUBE ASSEMBLY

NOTE:-
FOR APPLICABLE SPECIFICATIONS SEE MILITARY SPECIFICATION MIL-B-20278

DR 982515

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES ON FRACTIONS ± DECIMALS ± ANGLES DO NOT SCALE THIS DRAWING MATERIAL	APPROVED & RELEASED FOR PRODUCTION BY DIRECTION OF THE CHIEF OF ORDNANCE		FLOTATION BLADDER MK 1 MOD 1	DEPARTMENT OF THE NAVY BUREAU OF ORDNANCE WASHINGTON D. C.
	DRAFT CHK.	11-15		
	DESIGN HEAD	11-15		
	DESIGN ENGR.	11-15		
	SECTION HEAD	11-15		
	SECTION CHIEF	11-15		
	CHECKED		SCALE 1/4	RECORD DRAWING NO. 982515
	DRAWN	L.P.G.		

FIGURE 2

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